

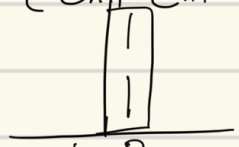
• In 2's complement addition/subtraction.

if the two numbers have the same sign bit and the sum has a different.

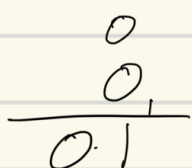
then overflow

$\Leftrightarrow V = C_n \oplus C_{n-1}$

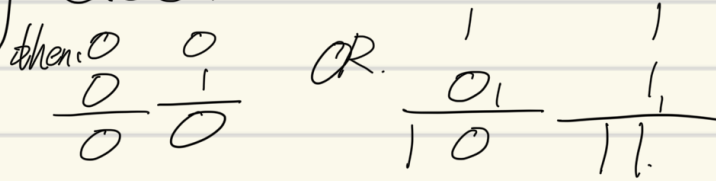
($C_n=1, C_{n-1}=0$ indicate:



$C_n=0, C_{n-1}=1$ indicate:

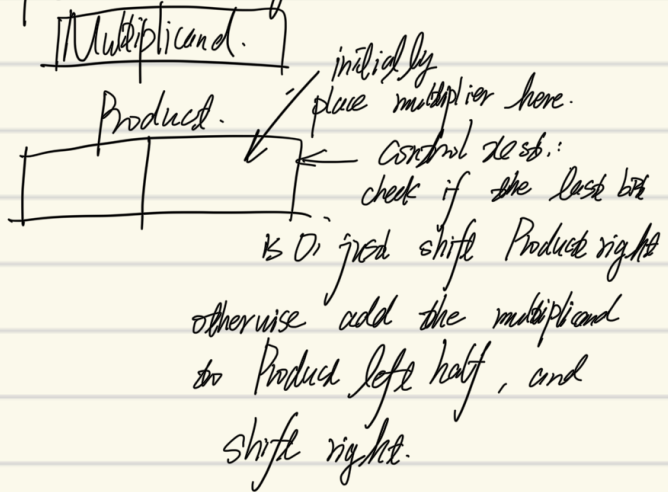


if $C_n \oplus C_{n-1} = 0$



No Overflow Here)

Multiplication: (Unsigned)



Signed Multiplexer.

Booth's Algorithm: last bit, former last bit

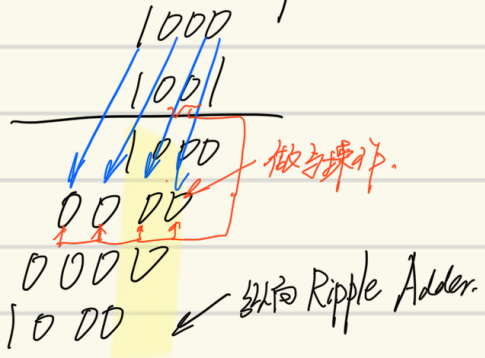
multiplexer:

00 11: Do Nothing.

01: ADD 10: Subtract

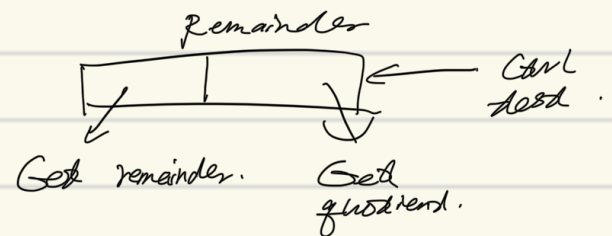
Initialize the former last bit to 0.

Matrix Multiplication:



• Division.

Divisor



First. Remainder Shift left.
Can avoid the initial meaning less subtraction.

loop: Remainder Divisor

if $\geq 0 \Rightarrow$ quotient 1.

$< 0 \Rightarrow$ produce quotient 0. do '+' last time.

Remainder Shift left

Finally. Remainder Shift right (Escape quotient).